

# FARZAN R S

+91 7356509774 ◊ Kayamkulam, Kerala, India

[farzanrichu610@gmail.com](mailto:farzanrichu610@gmail.com) ◊ [linkedin.com/in/farzan7](https://www.linkedin.com/in/farzan7) ◊ [github.com/f7z7](https://github.com/f7z7)

Electrical and Electronics Engineering student with strong foundations in Control Systems, Power Electronics, Instrumentation, and Automation. Hands-on experience in MATLAB/Simulink modeling, engineering software development, system analysis, and real-time data visualization using Python. Developed projects involving control-system design, simulation, IoT-enabled monitoring, and engineering data analysis. Strong problem-solving, communication, leadership, and teamwork skills with experience collaborating on technical projects and delivering engineering solutions in academic and professional environments.

## EDUCATION

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### Bachelor of Technology in Electrical and Electronics Engineering

2022 – 2026

Rajiv Gandhi Institute of Technology, Kottayam

CGPA: 8.67

**Relevant Coursework:** Control Systems, Power Electronics, Machine Learning, Signal Processing

### Higher Secondary Education

2020 – 2022

Govt Boys HSS Kayamkulam

Percentage: 95.5%

## TECHNICAL SKILLS

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|-----------------------------|----------------------------------------------------------------------------------------------|
| <b>Professional Skills</b>  | Problem Solving, Team Collaboration, Communication                                           |
| <b>Programming</b>          | MATLAB, Python, C++, C, JavaScript, C sharp                                                  |
| <b>System Modeling</b>      | MATLAB, Simulink, Stateflow                                                                  |
| <b>Core Knowledge</b>       | Control Systems, PID Controller Design, Performance Evaluation, Power Electronics Simulation |
| <b>Software &amp; Tools</b> | Git, GitHub, VS Code, Postman                                                                |

## PROJECTS

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### Solar Panel Fault Analysis and Monitoring System – Final Year Project

- Developed a monitoring and analysis system for photovoltaic installations using MATLAB/Simulink, machine learning, and IoT technologies.
- Simulated normal and abnormal operating conditions including shading, degradation, and sensor faults to support system investigation and fault identification.
- Generated and analyzed engineering datasets to identify fault conditions and evaluate system performance.
- Integrated ESP32-based communication for real-time monitoring and transmission of operational parameters.
- Developed visualization and reporting tools in Python to support engineering analysis and troubleshooting.
- Applied data-driven approaches to improve reliability and operational visibility.

### Temperature Controller ECU – MATLAB/Simulink Stateflow

[GitHub](#)

- Developed a temperature control Electronic Control Unit (ECU) using MATLAB/Simulink and Stateflow.
- Implemented state-based control logic to regulate temperature under different operating conditions.
- Modeled, simulated, and validated system behavior using Stateflow state machines and control strategies.

### Dual Converter Modeling – MATLAB/Simulink

- Developed a MATLAB/Simulink model of a dual converter to study bidirectional power flow operation.

- Analyzed converter performance in both rectification and inversion modes under different operating conditions.
- Evaluated voltage, current, and power characteristics to understand converter behavior and control requirements.

## EXPERIENCE

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**Python GUI Developer Intern**  
Nhanz Pvt. Ltd.

Jun 2025 – Jul 2025  
*Remote*

- Developed Python-based engineering software for real-time signal visualization and analysis.
- Built modular PyQt applications supporting multi-channel monitoring and data processing.
- Implemented data logging and diagnostic tools for engineering workflows.
- Participated in debugging, testing, and integration of software modules.
- Improved maintainability and scalability through modular software architecture.

**Webmaster**  
IEEE Student Branch RIT

Apr 2025 – Mar 2026  
*Kottayam, Kerala*

- Maintained and upgraded the official IEEE SB RIT website.
- Developed technical event websites using React and TypeScript.
- Collaborated with student teams to support digital operations and technical events.

## LEADERSHIP ACTIVITIES

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- Active volunteer in IEEE and ISTE technical activities and event coordination.
- Coordinated Quiz Club activities and organized academic and technical events.
- Worked with multidisciplinary teams to plan and execute technical initiatives.
- Demonstrated communication, collaboration, and organizational skills through student leadership activities.